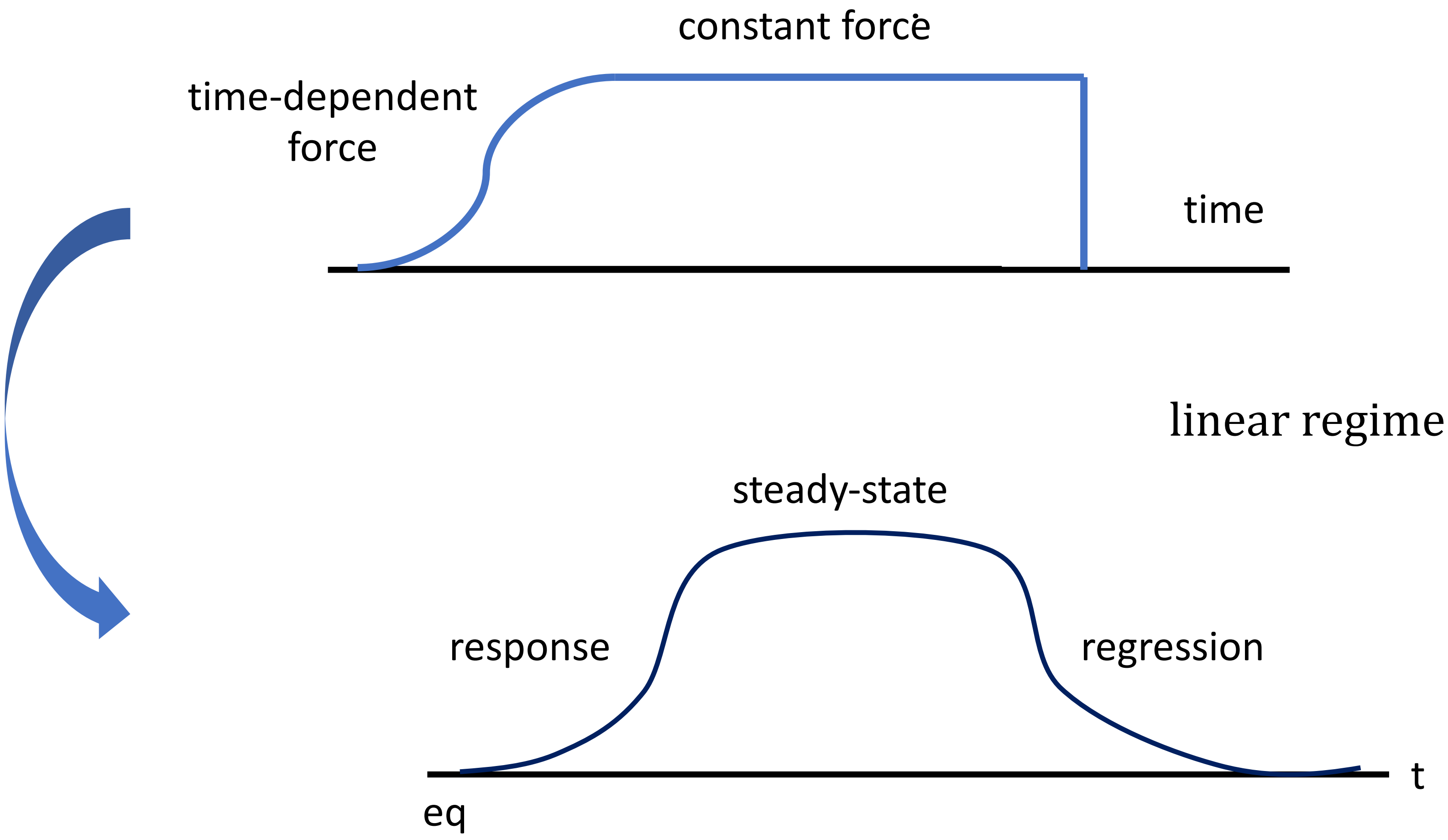
Lec. 2. Linear Regression and Response



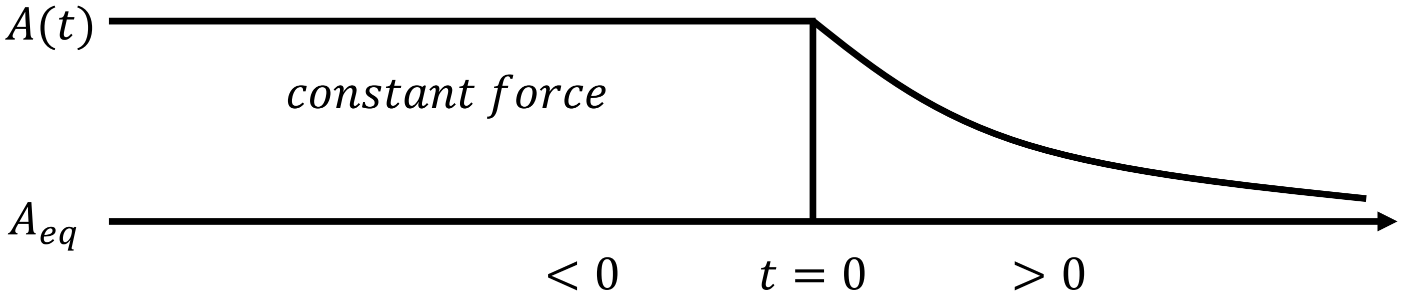
linear regime: $\delta A(\lambda f(t), t) = \lambda \delta A(f(t), t)$

$\delta A = \langle A \rangle_f - A_0$

2.1. Onsager regression theory

(1) Lars Onsager in 1931

2D Ising model
reciprocal relations
superfluidity
Onsager-Machlup functional
etc.



$$H_F = H - FA$$

$$\Delta A = \langle A \rangle - \langle A \rangle_{eq}$$

(2) Steady-State: displaced equilibrium

$A(t)$ —————————— $H_f = H - FA$

$constant\ force$

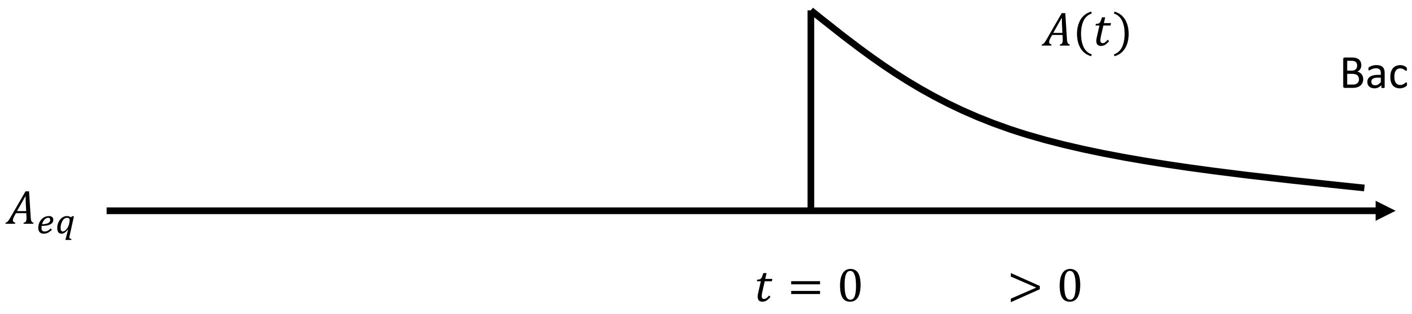
A_{eq} ——————————→

$$\langle A\rangle_F = \frac{\left\langle Ae^{-\beta(H-FA)}\right\rangle}{\left\langle e^{-\beta(H-FA)}\right\rangle} \sim \frac{TrAe^{-\beta H} + TrAA\beta Fe^{-\beta H}}{Tr\, e^{-\beta H} + Tr(\beta FA)e^{-\beta H}}$$

$$= \frac{\langle A\rangle + \langle AA\rangle\beta F}{1 + \beta F\langle A\rangle} \approx \langle A\rangle + [\langle AA\rangle - \langle A\rangle^2]\beta F$$

$$\therefore \Delta A_F = \langle A\rangle_F - \langle A\rangle \approx [\langle AA\rangle - \langle A^2\rangle]\beta F = C(0)\beta F$$

(3) time-dependence: regression

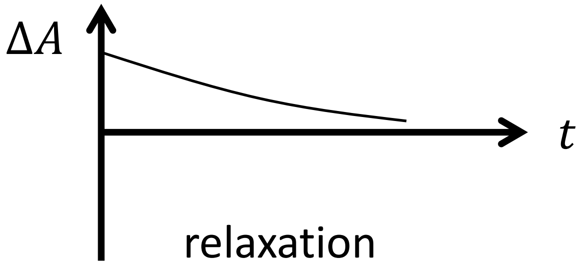


Back to equilibrium

$$A(t) = \frac{\langle A(t)e^{-\beta(H-FA)}\rangle}{\langle e^{-\beta(HA)}\rangle} \approx \langle A\rangle + [\langle A(t)A(0)\rangle - \langle A\rangle^2]\beta F$$

$$\boxed{\Delta A(t) = C(t)\beta F}$$

non-equilibrium



relaxation

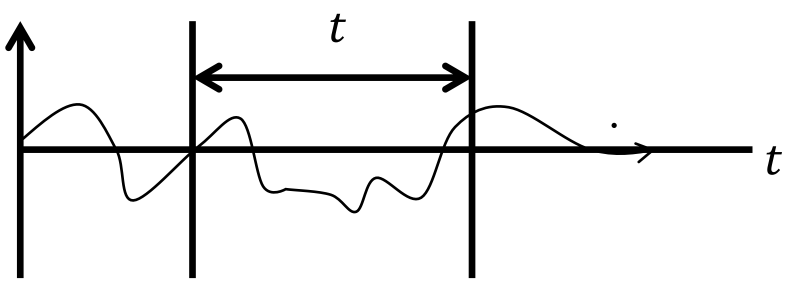
$$\Delta A = \langle A(t)\rangle - \langle A\rangle_{eq}$$

Non-eq

$$\frac{\Delta A(t)}{\Delta A(0)} = \frac{C(t)}{C(0)}$$

regression correlation

equilibrium



fluctuation

2.2 Auto-correlation function

$$C_{AA}(t,t') = \langle A(t)A(t')\rangle = \frac{\mathrm{Tr}[A(t)A(t')\rho_{eq}]}{\mathrm{Tr}[\rho_{eq}]}$$

1. Time invariance: C(t,t') = C(t-t')

2. $|\langle X|Y\rangle|^2 \le \langle X^2\rangle\langle Y^2\rangle$

   $$C^2(t) = |\langle A(t)|A(0)\rangle|^2 \le \langle A^2(t)A^2(0)\rangle \le \langle A^2(0)\rangle^2 = C^2(0)$$

   $$\therefore\ |C(t)| \le C(0)$$

3. $C = \langle A(0)A(-t)\rangle = C(-t) \Rightarrow$ Classical

   C is real and even. (not true for quantum systems)

4. $\lim_{t\to\infty}\langle A(t)A(0)\rangle = \langle A\rangle^2 \to 0$ since we define <A> = <δa> = 0

5. Ergodicity

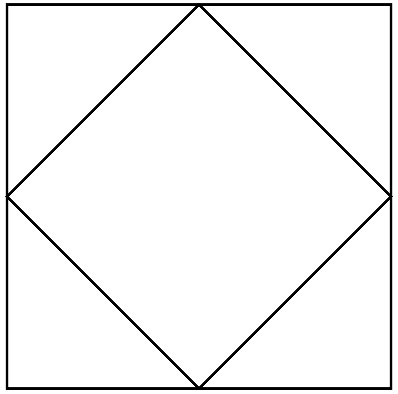
$$\boxed{\langle A(t)A(0)\rangle = \lim_{t\to\infty}\frac{1}{\tau}\int_0^{\tau} A(t+\tau')A(\tau')d\tau'}$$

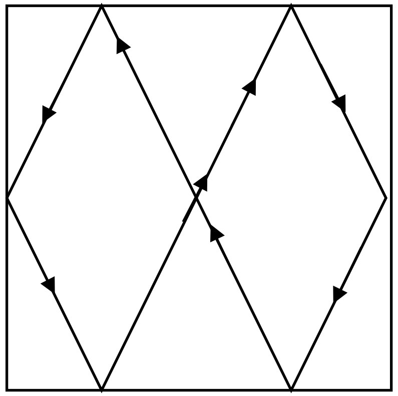
ensemble average time average

harmonic oscillator any harmonic motion → non-ergodic

two coupled Morse $\underbrace{\text{normal} \rightarrow \text{local}}_{\text{regular}} \rightarrow$ chaotic

billiard ball





rational
numbers

Quantum correlation function

(1) A(t) is in Heisenberg representation

C(t)=$\langle \hat{A}(t)\hat{A}(0)\rangle$. where $\acute{A}(t) = e^{iHt}\hat{A}e^{-iHt}$

(2) $C(-t) = C^*(t)$ $\quad$ $\mathrm{C(t)} = \mathrm{ReC} + \mathrm{i\ Im\ C}$

Re C $\rightarrow$ even $\quad$ Im C $\rightarrow$ odd

(3) Why not real?

$C(t)$ is not measurable, if A does not commute with H

(4) Detailed balance

$$\boxed{C(t - i\hbar\beta) = C(-t) = C^*(t)}$$

Example: Classical and Quantum LHO

formal solution

harmonic oscillator $\ddot{x} + \omega^2 x = 0$

Newton/Heisenberg equation are the same for LHO

$$\hat{x} = \hat{x}\cos(\omega t) + \frac{\hat{p}}{m\omega}\sin(\omega t)$$

$$\langle \hat{x}(t)\hat{x}(0)\rangle = \langle \hat{x}^2\rangle \cos\omega t + \langle \hat{p}\hat{x}\rangle \frac{\sin(\omega t)}{m\omega}$$

classical system

phase space average: <px> = <p><x> = 0

equal partition: <x^2> = $(\beta\omega^2 m)^{-1}$

$$\boxed{C(t) = \frac{kT}{m\omega^2}\cos\omega t}$$

Even function of time

Example: Classical and Quantum LHO

quantum dynamics

$$\langle \hat{x}(t)\hat{x}(0)\rangle = \langle \hat{x}^2\rangle \cos\omega t + \langle \hat{p}\hat{x}\rangle \frac{\sin(\omega t)}{m\omega}$$

Heisenberg uncertainty: [x, p] $= i\hbar$

2

$$\langle xp\rangle - \langle px\rangle = i\hbar \Rightarrow (xp) = \frac{\pm i\hbar}{2}$$

$$\langle x^2\rangle = \frac{\hbar}{2m\omega} coth\left(\frac{\alpha}{2}\right)$$

Quantum correlation

$$\boxed{\mathrm{C(t)} = \frac{\hbar}{2m\omega} coth\left(\frac{\alpha}{2}\right)\cos(\omega t) - i\,\frac{\hbar}{2m\omega}\sin(\omega t)}$$

Verify: even real part, odd imaginary part, and detailed balance

Example . Displaced Harmonic Osillator

initially displaced at $\bar{x}(0) = \dfrac{F}{m\omega^2}$

solution to EOM $\mathrm{m}\ddot{\mathrm{x}} + \mathrm{m}\omega^2 x = F(t)$

$$\langle x(t)\rangle = \underbrace{\langle x(0)\rangle \cos\omega t}_{= \frac{F}{m\omega^2}\cos\omega t} + \langle \dot{x}(0)\rangle \frac{\sin\omega t}{\omega}$$

previously $C(t) = \dfrac{kT}{m\omega^2}\cos(wt)$

thus $\boxed{\langle x(t)\rangle = \beta F C(t)}$ extend to damping

2.3. Linear Response Theory and Causality

$$C(t) \to \Delta A = \beta C(t)F \to \text{response} \quad K(t)$$

2.3.1 Kubo response theory

1. linearity $\Delta A(t) = \int_{-\infty}^{\infty} K(t,\tau)F(\tau)d\tau$

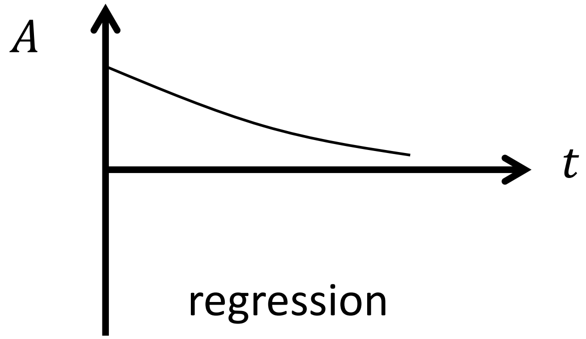
2. stationarity $\mathrm{K}(\mathrm{t},\tau) = \mathrm{K}(\mathrm{t}-\tau)$

3. causality $\int_{-\infty}^{\infty} \to \int_{-\infty}^{t} K\,(t-\tau)F(\tau)d\tau$

$$\boxed{\Delta A(t) = \int_{-\infty}^{t} K\,(t-\tau)F(\tau)d\tau}$$

4. Linear regression as a special of linear response

$$F(t) = \begin{cases} F & t > 0 \\ 0 & t < 0 \end{cases}$$



Linear regression: $\Delta A(t) = \beta C(t)F$

$$\Delta A(t) = \int_{-\infty}^{0} K(t-\tau)F d\tau = \beta C(t)F$$

Taking derivative on both sides of the equation, we obtain

$$\boxed{K(t) = -\beta\dot{C}(t)\theta(t)}$$

real and odd function, K(0)=0

Example: Linear Harmonic Oscillator (LHO)

1. Direct solution to the driven oscillator:

$$\ddot{x} + \omega^2 x = F(t)/m$$

Laplace transformation of EOM

.

$$(s^2 + \omega^2)\tilde{x}(s) = \tilde{F}(s)/m \Longrightarrow \tilde{x}(s) = \frac{\tilde{F}(s)}{m(s^2+\omega^2)}$$

$$x(t) = \int_{-\infty}^{t} \frac{\sin[\omega(t-\tau)]F(\tau)}{m} d\,\tau = \int_{-\infty}^{t} K(t-\tau)F(\tau)d\,\tau$$

2. Response theory

$$C(t) = \frac{k_B T}{m\omega^2}\cos\omega t \qquad K(t) = \frac{1}{m\omega}\sin(\omega t)\,\theta(t)$$

$\Longrightarrow$ no temperature-dependence
quantum response function = classical response function

2.4 Susceptibility and Absorption Spectrum

1. Susceptibility

$$\chi(\omega) = \int_0^{\infty} e^{i\omega t} K(t)dt$$
$$= \chi'(\omega) + i\chi''(\omega)$$

$$\Delta\tilde{A}(\omega) = \iint e^{i\omega(t-\tau)} e^{iw\ \tau} K(t-\tau)F(\tau)d\tau\, dt$$
$$= \int e^{i\omega(t-\tau)} K(t-\tau)d(t-\tau) \int e^{i\omega\tau} F(\tau)dt$$
$$= \chi(\omega)\tilde{F}(\omega)$$

$$\boxed{\Delta\tilde{A}(\omega) = \chi(\omega)\tilde{F}(\omega)}$$

2

$$\chi''(\omega) = \int_0^{\infty} \sin\omega t K(t)dt = \int_0^{\infty} \sin\omega t \left(-\beta\dot{C}(t)\right) dt$$
$$= \beta\omega \int_0^{\infty} C(t)\cos\omega t\, dt = \frac{\beta\omega}{2} \int_{-\infty}^{\infty} e^{i\omega t} C(t)dt = \frac{\beta\omega}{2} \tilde{C}(\omega)$$

$$\boxed{\chi'' = \frac{\omega}{2} C(\omega)}$$

3 Absorption spectrum

The rate at which work is done on the system

$$\int P(t)dt = \int F(t)\Delta\dot{A}(t)dt$$

$$= \frac{1}{2\pi}\int \tilde{F}(-i\omega)\Delta\tilde{A}(\omega)d\omega$$

$$= \frac{1}{2\pi}\int \tilde{F}(i\omega)\tilde{\chi}(\omega)\tilde{F}(-i\omega)d\omega$$

where $\Delta\tilde{A}(\omega) = \chi(\omega)\tilde{F}(\omega)$ and $\tilde{F}(-i\omega) = \hat{F}^*(-i\omega)$

$$\therefore \int P(\omega)d\omega\frac{1}{2\pi} = \frac{1}{2\pi}\int(-i\omega)\chi\left|\tilde{F}\right|^2 d\omega$$

$$= \frac{1}{2\pi}\int \omega\chi''\left|\tilde{F}\right|^2$$

$$\boxed{P(\omega) = \omega\chi''(\omega)\left|\tilde{F}(\omega)\right|^2}$$

4. Causality and the Kramers-Kronig relations

$$k(t) = 0 \quad \forall t < 0 \qquad \left| \int_0^{\infty} K(t)dt \right| < \infty$$

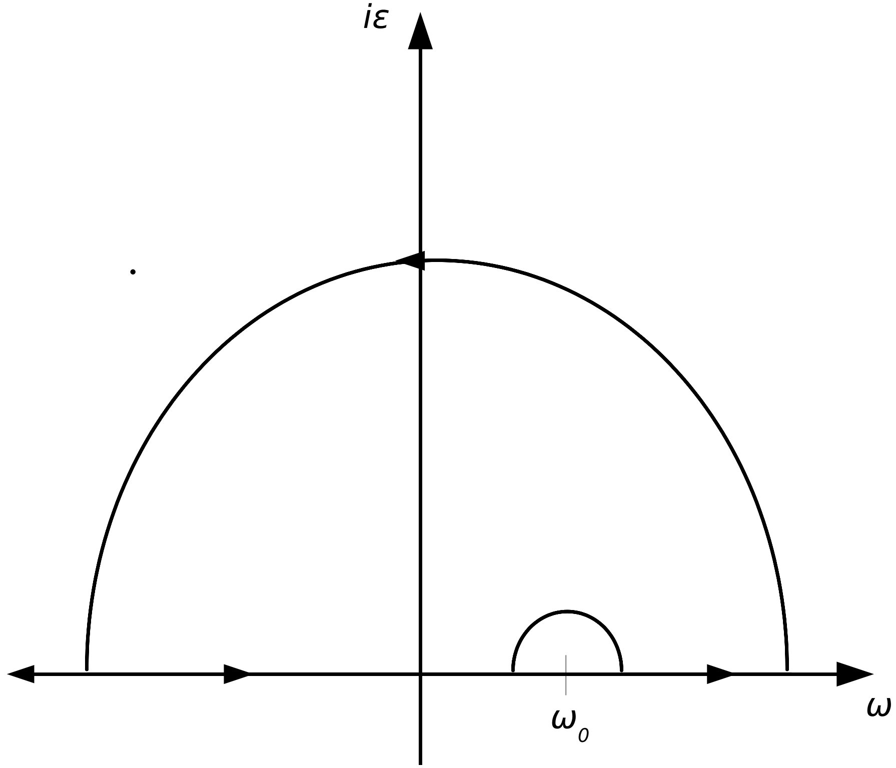
χ is analytic on the upper plane

$$\oint \frac{\chi(z)}{z - \omega_0} dz = \mathrm{P} \int_{-\infty}^{\infty} \frac{\chi(z)}{z - \omega_0} dz - i\pi\chi(\omega_0) = 0$$

$$\therefore \quad \mathrm{P} \int_{-\infty}^{\infty} \frac{\chi(z)}{z - \omega} dz = i\pi\chi(\omega)$$

$$\chi = \chi' + i\chi''$$

$$\boxed{\begin{aligned} \chi'(\omega) &= \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\chi''(z)}{z - \omega} dz \\ \chi''(\omega) &= -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\chi'(z)}{z - \omega} dz \end{aligned}}$$



contour integral

2.5. Quantum vs. classical response function

1. Classical and quantum response

$$k = \frac{i}{\hbar}\langle[A(t), A(t')]\rangle \qquad \text{or} \qquad \text{K(t)} = -\frac{2}{\hbar}\,\text{Im C(t)}$$

classical limit $\hbar \to 0$ $\qquad \lim_{\hbar - 0}\frac{1}{i\hbar}\underbrace{[A, B]}_{\text{commutator}} = \underbrace{\{A, B\}}_{\text{Poisson bracket}}$

$$\text{K(t)} = -\langle\{A(t), A(0)\}\rangle = -\frac{\text{Tr}\{e^{-\beta H}A(t)\}A(0)}{\text{Tr}\, e^{-\beta H}}$$

$$= \frac{\text{Tr}\,\beta\{A(t), H\}e^{-\beta H}A(0)}{\text{Tr}\, e^{-\beta H}} = -\beta\langle\dot{A}(t)A(0)\rangle = -\beta\dot{C}$$

2*. Divergence of micro-canonical response function

$$\mathrm{K}(t) = -\left\langle \frac{\partial A(t)}{\partial \Gamma(0)} \frac{\partial A(0)}{\partial \Gamma(0)} \right\rangle \Rightarrow \frac{\partial A(t)}{\partial p} \rightarrow \text{linear in t}$$

$$\frac{\partial q(t)}{\partial p(0)} \Rightarrow t \quad \text{long time} \qquad t \Rightarrow \infty$$

butterfly effect: instability of classical trajectories

3. High order

$$k^2 = \left(\frac{i}{\hbar}\right)^2 < \left[\left[\hat{A}(f_1 + f_2), \hat{A}(t_2)\right]\hat{A}(0)\right]\rangle$$

vanish for LHO, so nonlinear spectroscopy can probe anharmonicity

*PRL 95, 180405 (2005): PRL 96, 030403 (2006)